

JEE MAIN REVISION MASTER: HYDROCARBONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Alkanes & Alkenes: Additions & Named Reactions	15 PYQs	Hydrocarbons
Standard	Alkynes & Aromatic Electrophilic Substitution	0 PYQs	Alkynes & Aromatics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Alkanes & Alkenes: Additions & Named Reactions	$\text{Wurtz: } 2\text{RX} + 2\text{Na} \rightarrow \text{dry ether} \text{ R-R} + 2\text{NaX}$ $\text{Ozonolysis: } >\text{C}=\text{C}< \rightarrow \text{O}_3, \text{Zn/H}_2\text{O} \rightarrow \text{C=O} + \text{O=C}<$
Alkynes & Aromatic Electrophilic Substitution	$\text{Acidity: } \text{CH} \equiv \text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3\text{CH}_2\text{CH}_3$ $\text{Nitration Electrophile: } \text{NO}_2^+$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Alkanes & Alkenes: Additions & Named Reactions (Very High Yield — 15 PYQs)

Core Principle: Wurtz reaction couples alkyl halides to form symmetrical alkanes. Markovnikov addition: electrophilic proton adds to carbon with more hydrogens. Anti-Markovnikov (peroxide effect) occurs ONLY with HBr in the presence of peroxides via free radical mechanism. Ozonolysis cleaves $>\text{C}=\text{C}<$ into aldehydes and ketones.

■ **Caution / JEE Pitfall:** The peroxide effect applies strictly to HBr; it DOES NOT work with HCl or HI due to thermodynamic and kinetic bond enthalpy constraints.

■ **Speed Trick:** Ozonolysis products directly reveal the position of the double bond: replace $>\text{C}=\text{O}$ and $\text{O}=\text{C}<$ with a double bond to reconstruct the starting alkene.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The major product formed when 1-Bromo -3- Chlorocyclobutane reacts with metallic sodium in dry ether is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "2-methyl propane can be prepared by Wurtz reaction. The haloalkanes taken along with metallic sodium and dry ether are :..." [Ans: N/A]

3.2 Alkynes & Aromatic Electrophilic Substitution (Standard Yield — 0 PYQs)

Core Principle: Acidity of terminal alkynes: sp -hybridized carbon has 50% s -character, making terminal hydrogen acidic enough to react with NaNH_2 or ammoniacal AgNO_3 . Hydration of acetylene with $\text{Hg}^{2+}/\text{H}_2\text{SO}_4$ gives acetaldehyde; propyne gives acetone. Electrophilic aromatic substitution on benzene: Nitration (electrophile NO_2^+ generated by $\text{HNO}_3 + \text{H}_2\text{SO}_4$), Friedel-Crafts alkylation/acylation with anhydrous AlCl_3 .

■ **Caution / JEE Pitfall:** Aniline cannot undergo Friedel-Crafts reaction because its basic $-\text{NH}_2$ group coordinates with the Lewis acid catalyst AlCl_3 to form an inactive salt complex.

■ **Speed Trick:** Terminal alkynes give a red precipitate of copper acetylide with ammoniacal cuprous chloride, distinguishing them from non-terminal alkynes.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "The major product formed when 1-Bromo -3- Chlorocyclobutane reacts with metallic sodium in dry ether is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered conformation is more. Read the above statements and choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are false
- (B) Both Statement I and Statement II are true
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement I is true. In ethane, the staggered conformation is more stable because the hydrogen atoms on the two carbon atoms are as far apart as possible. The dihedral angle between C-H bonds is 60° , so the repulsion between bonding electron pairs is minimum. Statement II is false. Torsional strain is not more in the staggered conformation. It is maximum in the eclipsed conformation because the C-H bonds of the two carbon atoms are directly aligned with each other. This causes greater repulsion between bonding electron pairs. So, $\text{Statement I is true but Statement II is false.}$